

Quantum molecular dynamics

9.1 Introduction

In the previous chapter we considered systems of interacting particles. They were treated as classical particles for which the interaction potential is known. We had to solve the classical equations of motion to simulate the behaviour of such a system at some nonzero temperature. Had we added frictional forces, the system would have evolved towards the ground state. In this chapter we discuss methods for simulating interacting atoms and molecules using quantum mechanical calculations. In fact, we consider the nuclei on a classical level but use quantum mechanics for the electronic degrees of freedom. Again, we can use this approach either to simulate a system of interacting particles at a finite temperature, or to find the ground state (minimum energy) configurations of solids and of molecules.

In [Chapters 4 to 6](#) we studied methods for solving the electronic structure of molecular and solid state systems with a static configuration of nuclei (Born–Oppenheimer approximation). Knowledge of the electronic structure includes knowledge of the total energy. Therefore, by varying the positions of the nuclei, we can study the dependence of the total energy on these positions. The energy $E(\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_N)$ as a function of the nuclear positions $\mathbf{R}_i$ is called the *potential surface*. As a simple example, consider the hydrogen molecule. We assume that the molecule is not rotating, so that the nuclear motion is a vibration along the molecular axis. The only relevant parameter describing the relative positions of the two nuclei is their separation X . The force between the nuclei is then given by $F = -\partial E(X)/\partial X$ (see [Figure 9.1](#)). These forces are usually parametrised and the parameters are fixed by comparison with quantum mechanical calculations for a few configurations, or by comparison with experimental results. This parametrised form can then be used to calculate the motion of the nuclei on a classical level, for example to find the equilibrium conformation of the molecule, which is the configuration of nuclei that minimises the total energy. This is called the method of *force fields*; it is often used by chemists.

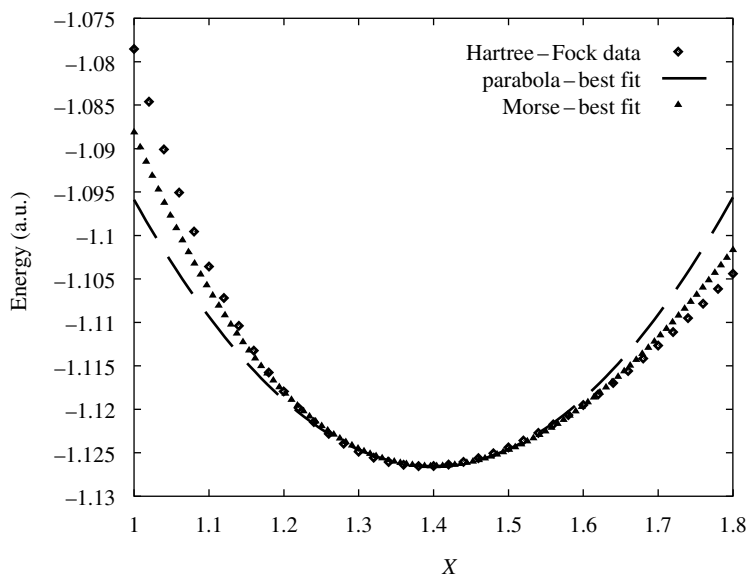


Figure 9.1. The effective potential of the hydrogen nuclei in the hydrogen molecule as a function of the separation X . A harmonic (parabolic) potential and a Morse potential are fitted to the bottom of the well.

The parametrisation of the forces is often carried out for small deviations of the configuration from the equilibrium conformation, so that the potential energy can be approximated quite accurately by harmonic potentials, such as stretching, bending, and torsional potentials, encountered in [Section 8.6.1](#). The motion can then be decomposed into normal modes, by defining new coordinates in terms of which the system can be described as a collection of uncoupled harmonic oscillators. This problem then has an analytic quantum mechanical solution, leading to discretised energy levels which can be compared with experiment. So, although the force field method treats the nuclear motion classically, we can obtain quantum mechanical solutions for the *nuclear* motion from it (within a Born–Oppenheimer approach).

In our example of the hydrogen molecule we can fit the bottom of the potential well shown in [Figure 9.1](#) by a harmonic potential. Since the well is rather asymmetric, a more reliable fit is provided by the Morse potential, for which the spectrum is also known analytically (see Problem A4). For the harmonic approximation $\kappa X^2/2$, the angular frequency $\omega = \sqrt{\kappa/m}$ and the spectral levels are given as $\hbar\omega(n + 1/2)$. For the hydrogen molecule, the mass to be used is the reduced mass, which is about half the proton mass (i.e. 918.8 electron masses if we neglect the mass of the two electrons), and we find $\kappa = 0.3850$ (in atomic units) so that the frequency

becomes $\nu_{\text{vibr}} = 13.64 \times 10^{13}$ Hz, to be compared with the experimental value $\nu_{\text{vibr}} = 12.48 \times 10^{13}$ Hz [1].¹

The harmonic approximation works well for low energies. It is used for stretch, bend and torsional energies; see also [Section 8.6.1](#). For higher energies, anharmonic terms can be included in the potentials – see the Morse potential in the hydrogen example. For energies much higher than the spacing between the energy levels, quantum effects do not affect physical properties and a fully classical description is appropriate.

For interacting molecules the force field procedure becomes unfeasible because we would have to calculate energies and variations in energies for all possible relative positions and orientations for pairs or sets of two and more molecules, which becomes exceedingly tedious and (computer) time-consuming. Therefore, in these systems, the intramolecular interactions are modelled by force fields and the intermolecular interactions by atomic pair-interactions as we have seen throughout the previous chapter. Although this approach yields rather accurate results, in particular when the density is not too high, the use of these pair-potentials is not justified for dense systems. Moreover, the energy differences between different atomic conformations are often very small, so that high accuracy is needed for predicting equilibrium structures.

To achieve accuracy in these cases, it is necessary to calculate forces and energies from quantum electronic structure calculations; if this is unfeasible for all possible configurations, we take the more economical approach of calculating forces and energies only for those configurations which actually occur in the simulation. We must therefore perform an electronic structure calculation at every molecular dynamics time step, and derive the *force* on the nuclei from that calculation. The word *force* is emphasised because the methods described in the first few chapters of this book aimed at calculating the energies and not the forces. Of course, it would be possible to derive the forces from the energies by studying the variations in the latter with nuclear positions but that would require an exceedingly large number of energy calculations. It is better therefore to try to find methods for calculating the forces directly.

The energy of a system of electrons in its ground state ψ_G for a fixed configuration of nuclei $S = (\mathbf{R}_1, \dots, \mathbf{R}_M)$, where $\mathbf{R}_n$ is the position of the n th nucleus, is given by²

$$E = \frac{\langle \psi_G | H(S) | \psi_G \rangle}{\langle \psi_G | \psi_G \rangle}. \quad (9.1)$$

¹ In atomic units, the unit of frequency is $\alpha c/a_0 = 4.13414 \times 10^{14}$ Hz; α is the fine structure constant.

² We use the letter S in order to avoid confusion with $R = (\mathbf{r}_1, \dots, \mathbf{r}_N)$.

The (classical) force on nucleus n is given as the negative gradient ∇_n of the energy with respect to the $\mathbf{R}_n$:

$$\mathbf{F}_n = -\nabla_n E(S) = -\nabla_n \left[\frac{\langle \psi_G | H(S) | \psi_G \rangle}{\langle \psi_G | \psi_G \rangle} \right]. \quad (9.2)$$

It should be noted that there is not only the explicit S -dependence in the Hamiltonian, but the ground state is evaluated for the Hamiltonian with a particular configuration S . Therefore the ground state also depends on S .

The *Hellmann–Feynman theorem* [2, 3], which we discussed for the single-parameter case in [Section 5.3](#), states that we can neglect this dependence: if ψ_G is an eigenstate of the Hamiltonian $H(S)$, we have

$$\begin{aligned} & (\langle \psi_G | \psi_G \rangle)^2 \nabla_n E \\ &= \left[\langle (\nabla_n \psi_G) | H | \psi_G \rangle + \langle \psi_G | (\nabla_n H) | \psi_G \rangle + \langle \psi_G | H | (\nabla_n \psi_G) \rangle \right] \langle \psi_G | \psi_G \rangle \\ &\quad - \langle \psi_G | H | \psi_G \rangle \left[\langle (\nabla_n \psi_G) | \psi_G \rangle + \langle \psi_G | (\nabla_n \psi_G) \rangle \right], \end{aligned} \quad (9.3)$$

where we have omitted the S -dependence of the Hamiltonian. Except for the term including $\langle \psi_G | (\nabla_n H) | \psi_G \rangle$, all the terms on the right hand side cancel: this follows directly from the fact that $H\psi_G = E_G\psi_G$ and from the fact that H is Hermitian. So we are left with

$$\nabla_n E = \frac{\langle \psi_G | (\nabla_n H) | \psi_G \rangle}{\langle \psi_G | \psi_G \rangle}. \quad (9.4)$$

In practice we do not know the exact ground state, but we have only a variational approximation to it. Therefore, in actual calculations, the Hellmann–Feynman theorem does not predict the actual forces exactly and the variation of the (approximated) ground state wave function should be taken into account as well. Nevertheless, the Hellmann–Feynman theorem is used quite often for predicting ground state configurations, because the inclusion of other contributions is cumbersome.

9.2 The molecular dynamics method

In principle, all the ingredients for a molecular dynamics simulation using forces calculated from the quantum electronic structure are at our disposal. However, at each step in the MD simulation, a full electronic structure calculation is required, so the method consumes a lot of computer time. In 1985, Car and Parrinello proposed a method in which not only the nuclear positions, but also the electronic states are calculated using MD algorithms. This results in a description of the system in which the electronic structure does not, in general, relax completely to the ground state of the actual configuration of nuclei; however, the calculated electronic structure will follow the exact one rather closely. We start the description of the Car–Parrinello

method by recalling the energy functionals of the Hartree–Fock and the density functional theory (see [Chapters 4 and 5](#)).

The ground state Hartree–Fock wave function for N electrons can be written as the Slater determinant

$$\Psi_G(R) = \det[\psi_k(\mathbf{x}_i)] = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\mathbf{x}_1) & \psi_2(\mathbf{x}_1) & \cdots & \psi_N(\mathbf{x}_1) \\ \psi_1(\mathbf{x}_2) & \psi_2(\mathbf{x}_2) & \cdots & \psi_N(\mathbf{x}_2) \\ \vdots & \vdots & & \vdots \\ \psi_1(\mathbf{x}_N) & \psi_2(\mathbf{x}_N) & \cdots & \psi_N(\mathbf{x}_N) \end{vmatrix}, \quad (9.5)$$

where the ψ_k are one-electron spin-orbitals; $\mathbf{x}_i$ is the combined spin and orbital coordinate of particle i . The spin-orbitals should satisfy the orthonormality requirements

$$\langle \psi_k | \psi_l \rangle = \delta_{kl}. \quad (9.6)$$

The energy is given in terms of the ψ_k as

$$E_{\text{HF}} = \sum_k \langle \psi_k | h | \psi_k \rangle + \frac{1}{2} \sum_{kl} [\langle \psi_k \psi_l | g | \psi_k \psi_l \rangle - \langle \psi_k \psi_l | g | \psi_l \psi_k \rangle]. \quad (9.7)$$

h is the one-electron Hamiltonian and g is the electron–electron Coulomb repulsion (see [Chapter 4](#)). Minimisation of this expression with respect to the ψ_k subject to the constraint (9.6) requires the Fock equation to be satisfied:

$$\mathcal{F}\psi_k = \sum_l \Lambda_{kl} \psi_l \quad (9.8)$$

with

$$\begin{aligned} \mathcal{F}\psi_k = & \left[-\frac{1}{2} \nabla^2 - \sum_n \frac{Z_n}{|\mathbf{r} - \mathbf{R}_n|} \right] \psi_k(\mathbf{x}) + \sum_{l=1}^N \int d\mathbf{x}' |\psi_l(\mathbf{x}')|^2 \frac{1}{|\mathbf{r} - \mathbf{r}'|} \psi_k(\mathbf{x}) \\ & - \sum_{l=1}^N \int d\mathbf{x}' \psi_l^*(\mathbf{x}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \psi_k(\mathbf{x}') \psi_l(\mathbf{x}). \end{aligned} \quad (9.9)$$

After a unitary transformation of the set $\{\psi_k\}$ (see [Section 4.5.2](#) and [Problem 4.7](#)), [Eq. \(9.8\)](#) transforms into

$$\mathcal{F}\psi_k = \varepsilon_k \psi_k. \quad (9.10)$$

Using $\mathcal{F}\psi_k = \delta E_{\text{HF}} / \delta \psi_k$, we can rewrite this as

$$\frac{\delta E_{\text{HF}}}{\delta \psi_k(\mathbf{x})} = \varepsilon_k \psi_k(\mathbf{x}). \quad (9.11)$$

The eigenvalues ε_k are the Fock levels; the energy can be calculated from these.

In density functional theory the energy can be written as a function of the ground state density, which in turn is written in terms of the basis functions as

$$n(\mathbf{r}) = \sum_{k=1}^N |\psi_k(\mathbf{r})|^2, \quad (9.12)$$

assuming that the states are ordered according to increasing energy. We have seen already in [Chapter 5](#) that there is no direct expression of the total energy as a function of the density, as the form of the kinetic energy functional of the density is unknown. The energy can however be obtained indirectly via the solutions ψ_k of the Kohn–Sham equations:

$$-\frac{1}{2}\nabla^2\psi_k(\mathbf{r}) + V_{\text{eff}}(\mathbf{r})\psi_k(\mathbf{r}) = \varepsilon_k\psi_k(\mathbf{r}) \quad (9.13)$$

where

$$V_{\text{eff}}(\mathbf{r}) = V_{\text{ion}}(\mathbf{r}) + \int d^3r' \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + V_{\text{xc}}[n](\mathbf{r}). \quad (9.14)$$

The exchange correlation potential V_{xc} on the right hand side is the derivative of the exchange correlation energy E_{xc} with respect to $n(\mathbf{r})$.

In terms of the ψ_k , the total DFT energy is written as

$$\begin{aligned} E_{\text{DFT}} = & - \sum_k \frac{1}{2} \langle \psi_k | \nabla^2 | \psi_k \rangle + \sum_k \langle \psi_k | V_{\text{ion}} | \psi_k \rangle \\ & + \frac{1}{2} \int d^3r d^3r' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + E_{\text{xc}}[n](\mathbf{r}). \end{aligned} \quad (9.15)$$

The Kohn–Sham equations can be written as

$$\frac{\delta E_{\text{DFT}}}{\delta \psi_k(\mathbf{r})} = \varepsilon_k \psi_k(\mathbf{r}), \quad (9.16)$$

i.e. the same form as [\(9.11\)](#).

Summarising, the total energy, which is the electronic energy (either E_{DFT} or E_{HF}) plus the electrostatic energy of the nuclei, can be written as a functional depending on the orbitals ψ_k and of the nuclear coordinates, collected together in the variable S :

$$E_{\text{tot}} = E_{\text{tot}}(\{\psi_k\}, S), \quad (9.17)$$

where the orbitals ψ_k form an orthonormal set. In both the Hartree–Fock and the density functional theory approach we minimise this energy with respect to the orbitals ψ_k , according to the variational principle. Usually, a finite basis set $\{\chi_r\}$ is used, in terms of which the orbitals are given as

$$\psi_k(\mathbf{r}) = \sum_r C_{rk} \chi_r(\mathbf{r}), \quad (9.18)$$

so that the energy can be written in terms of the C_{rk} and S :

$$E_{\text{tot}} = E_{\text{tot}}(\{C_{rk}\}, S). \quad (9.19)$$

As the basis functions are often centred on the atomic nuclei, they may contain an explicit S -dependence. Car and Parrinello used the form (9.17) (or (9.19)) with the constraint (9.6) as a starting point for finding the equilibrium conformation (i.e. the minimal energy conformation) by locating the minimum of the total energy as a function of the ψ_k (or the C_{rk}) and the nuclear coordinates S . This means that the electronic structure does not have to be calculated exactly for each conformation of nuclei, as both the electronic orbitals and the nuclear positions are varied *simultaneously* in order to locate the minimum.

The minimisation problem of the energy can now be considered as an abstract numerical problem, and any minimisation algorithm can in principle be applied. One possible approach is the simulated annealing method [4], which requires only the energy to be calculated – no force calculations are needed. However, Car and Parrinello assigned, aside from the time dependence of the nuclear coordinates, a *fictitious* time dependence to the electronic wave functions (or, in a linear variational calculation, the C_{rk}), and constructed a dynamical Lagrangian including the electronic wave functions and the nuclear coordinate S with their time derivatives as the variables. This leads to a classical mechanics problem with the energy (9.17) acting as a potential. If a friction term is then added to the equations of motion of this classical system, the degrees of freedom will come to rest after some time, with values corresponding to the minimum of the classical potential, which is the energy of the quantum system at the equilibrium configuration of the nuclei. It is also possible to put the frictional force equal to zero in order to simulate the system at a nonzero temperature.

The Lagrangian of the classical system reads

$$L(\{\psi_k\}, S) = \frac{\mu}{2} \sum_k \dot{\psi}_k^2 + \sum_n \frac{M_n}{2} \dot{\mathbf{R}}_n^2 - E_{\text{tot}}(\psi_k, S) + \sum_{kl} \Lambda_{kl} \langle \psi_k | \psi_l \rangle. \quad (9.20)$$

μ is some small mass (see below), and M_n is the actual mass of the n th nucleus, with position $\mathbf{R}_n$. The last term on the right hand side is necessary to ensure orthonormality of the ψ_k ; the Λ_{kl} must always be calculated from this requirement. Car and Parrinello suggested that this Lagrangian can be used not only for finding the minimum of the total energy, but also for performing real molecular dynamics simulations at finite temperature. It will be clear that in general, when the nuclei move, the method might not have produced the minimal energy of the electrons before the next nuclear displacement: the calculated electronic structure will ‘lag behind’ the nuclear motion. Although this retardation effect will occur in reality (the Born–Oppenheimer approximation neglects the fact that the electrons do not

have the opportunity to adapt themselves to the changing nuclear configuration at any time), there is no reason to believe that the retardation effect implied by the Car–Parrinello Lagrangian is related to real physical behaviour.

The details of the kinetic energy of the electrons do not matter: what matters is the fact that the mass μ used in this kinetic energy should be small enough to enable the electronic wave function to adapt reasonably well to the changing nuclear configurations. This mass should therefore be much smaller than the nuclear masses. The choice of the mass μ is determined by a trade-off between accuracy and efficiency. If we include friction in the equations of motion, the particular values of neither electronic nor nuclear masses matter, as we shall always end up with zero kinetic energy, at the minimum of the total energy of the system (which is the potential of the Car–Parrinello Lagrangian), although different choices of these masses lead to different rates of convergence towards the energy minimum.

Let us write down the equations of motion for the Car–Parrinello Lagrangian. We must take the orthogonality constraint (9.6) into account using Lagrange parameters $\Lambda_{kl}(t)$. The Euler–Lagrange equations now read

$$\mu \ddot{\psi}_k = -\frac{\partial E_{\text{tot}}}{\partial \psi_k} + 2 \sum_l \Lambda_{kl} \psi_l(\mathbf{r}) \quad (9.21)$$

and

$$M_n \ddot{\mathbf{R}}_n = -\frac{\partial E_{\text{tot}}}{\partial \mathbf{R}_n} + \sum_{kl} \Lambda_{kl} \frac{\partial \langle \psi_k | \psi_l \rangle}{\partial \mathbf{R}_n}. \quad (9.22)$$

The last term on the right hand side of the last equation vanishes if the basis functions do not depend on the nuclear positions S . As we know the total energy in both DFT and HF in terms of the orbitals ψ_k and $\mathbf{R}_n$, the energy derivatives occurring in these equations can be evaluated – see the next section.

Instead of assigning a kinetic energy to the orbitals ψ_k , leading to Eq. (9.21), we can assign a kinetic energy to the expansion coefficients C_{rk} . In that case, Eq. (9.21) becomes

$$\mu \ddot{C}_{rk} = -\frac{\partial E_{\text{tot}}}{\partial C_{rk}} + 2 \sum_l \Lambda_{kl} \sum_s S_{rs} C_{sl}. \quad (9.23)$$

If μ is allowed to depend on r and k , this equation can be made equivalent to (9.21) but, as argued above, the details of the kinetic energy do not matter that much as long as the electronic degrees of freedom can adapt themselves to the nuclear positions.

If a frictional term is added to the equations of motion, the solution will become stationary after some time, and the left hand side vanishes. Equation (9.21) then becomes an equation similar to the Fock and the Kohn–Sham equations ((9.11) and (9.16)), except for the eigenvalues ε_k being replaced by the matrix elements Λ_{kl} . This is precisely the same difference as we have encountered in the diagonalisation

of the Fock matrix (see [Section 4.5.2](#) and above): for $\ddot{\psi}_k = 0$, [Eq. \(9.21\)](#) reduces to an eigenvalue equation after an appropriate unitary transformation of the set $\{\psi_k\}$ and of the Lagrange parameters Λ_{kl} .

The values of the Lagrange parameters Λ_{kl} depend on time: they must be calculated at each MD step such that they guarantee the orthonormality constraint [\(9.6\)](#). This calculational procedure is related to the particular integration scheme used (the Verlet algorithm in our case). In [Section 8.6.2](#) we have encountered this problem already. Car and Parrinello have used the iterative SHAKE algorithm of Ryckaert *et al.* [\[5\]](#) (see [Section 8.6.3](#)) to solve for the Λ_{kl} . We return to the problem of calculating the Λ_{kl} in more detail below.

If the nuclear equilibrium configuration is searched for, starting from an initial configuration which might be far off the equilibrium, we are likely to end up in a local energy minimum instead of the global minimum. In this case, we might use the simulated annealing method [\[4\]](#) which allows the system to hop over local energy barriers to arrive at the global minimum.

It is interesting to compare the equations obtained here with the time-dependent Hartree–Fock (TDHF) equations. These are obtained from a variational treatment of the time-dependent Schrödinger equation using Slater determinants constructed from time-dependent spin-orbitals. The time-dependent Schrödinger equation can be derived as the stationarity condition of the functional

$$S = \int dt \int dX \Psi^*(X, t) \left(i\hbar \frac{\partial}{\partial t} - H \right) \Psi(X, t) \quad (9.24)$$

with $X = (\mathbf{x}_1, \dots, \mathbf{x}_N)$. By taking for $\Psi(R, t)$ a Slater determinant with time-dependent orbitals $\psi_k(\mathbf{x}, t)$, the stationarity condition leads to the following equation for the spin-orbitals [\[6\]](#):

$$i\hbar \frac{\partial}{\partial t} \psi_k(\mathbf{x}, t) = \mathcal{F} \psi_k(\mathbf{x}, t). \quad (9.25)$$

The TDHF equations lead to a conservation law for the overlap matrix $S_{kl}(t) = \langle \psi_k(t) | \psi_l(t) \rangle$. Hence, if we choose an orthonormal set to start off with at $t = 0$, the set will remain orthonormal in the course of time.

In comparison with the MD equation of motion for the electrons, [Eq. \(9.21\)](#), we see that the second derivative with respect to time is replaced by a first order one, and that there is no Lagrange parameter as a result of the overlap matrix being conserved.

Time-dependent Hartree–Fock is used for studying the quantum dynamics of scattering processes, for example in nuclear physics and in studies of scattering of electrons from atoms.

9.3 An example: quantum molecular dynamics for the hydrogen molecule

In this subsection we work out an application of the Car–Parrinello method to the hydrogen molecule in some detail. Our example is based on the Hartree–Fock calculation of the hydrogen molecule considered in [Chapter 4](#), in particular Problem 4.9. There are two spin-orbitals with opposite spin and the same orbital part. Therefore, the wave function is completely specified by the form of this orbital. We use the GTO basis set of Problem 4.9 with eight basis s-functions χ_r , four on each atom.

The molecular dynamics method can be restricted to the electronic structure part of the total energy, keeping the nuclear positions fixed. We do this first; later we shall extend the method to include nuclear displacements.

9.3.1 The electronic structure

The energy can be written as

$$E_{\text{tot}} = 2 \sum_{rs} C_r h_{rs} C_s + \sum_{rstu} C_r C_s C_t C_u \langle rt | g | su \rangle + \frac{1}{X}. \quad (9.26)$$

Note that there is no index k as the two electrons occupy only one orbital. The Fock matrix $\mathbf{F}$ is given by

$$F_{rs} = h_{rs} + \sum_{tu} C_t C_u \langle rt | g | su \rangle \quad (9.27)$$

(all sums over indices r, s, t, u run over the basis states, so in our case from 1 to 8). The normalisation condition for the orbital is

$$\sum_{rs} C_r S_{rs} C_s = 1. \quad (9.28)$$

Therefore, the equation of motion for the C_r (without friction) is given by

$$\begin{aligned} \frac{\mu}{4} \ddot{C}_r &= - \sum_s h_{rs} C_s - \sum_{stu} C_s C_t C_u \langle rt | g | su \rangle - \lambda \sum_s S_{rs} C_s \\ &= - \sum_s (F_{rs} + \lambda S_{rs}) C_s. \end{aligned} \quad (9.29)$$

We shall use the Verlet algorithm for solving the equations of motion. In this form, they read for $\mu = 4$:

$$C_r(t+h) = 2C_r(t) - C_r(t-h) - h^2 \sum_s (F_{rs} + \lambda S_{rs}) C_s(t). \quad (9.30)$$

Suppose we know the $C_r(t)$ and the $C_r(t-h)$. The solution to the equation of motion proceeds in two stages. First we calculate

$$\tilde{C}_r(t+h) = 2C_r(t) - C_r(t-h) - h^2 \sum_s F_{rs} C_s(t). \quad (9.31)$$

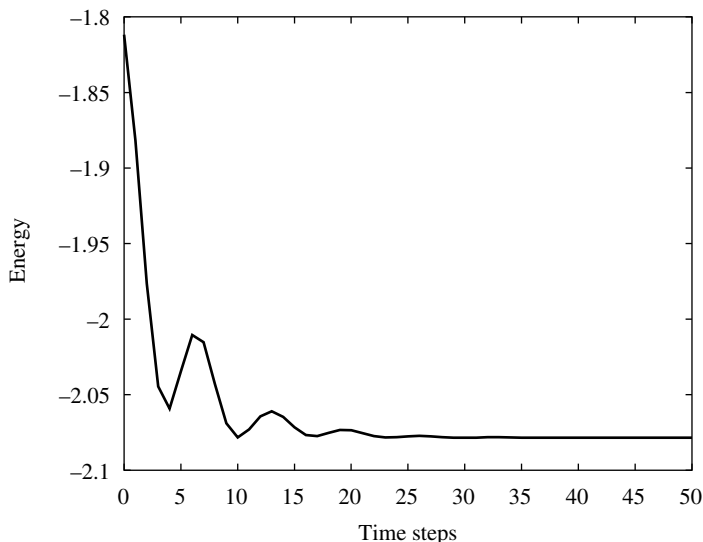


Figure 9.2. Evolution of the energy in a Car–Parrinello simulation of the electronic structure of the hydrogen molecule with separation $X = 1$ between the nuclei, with frictional forces included.

Now we must add an amount $-\lambda S_{rs}C_s(t)$ to this solution, where λ is determined by the requirement that the normalisation condition (9.28) holds:

$$\begin{aligned} \sum_{rs} \tilde{C}_r(t+h) S_{rs} \tilde{C}_s(t+h) - 2\lambda \sum_{rst} S_{rs} \tilde{C}_r(t+h) S_{rt} C_s(t) \\ + \lambda^2 \sum_{rstu} S_{rs} C_s(t) S_{rt} S_{tu} C_u(t) = 1. \end{aligned} \quad (9.32)$$

This is a quadratic equation in λ , of which the lowest positive root is needed. The Verlet solution of the equation of motion is now fully defined.

Modifying the HF program of Chapter 4 to calculate the electronic structure is relatively easy, as the Fock matrix and the overlap matrix are calculated already in this program.

PROGRAMMING EXERCISE

Take the program of Problem 4.9 and replace the self-consistency iteration by a molecular dynamics algorithm with friction, using the Verlet algorithm.

A frictional force $-\gamma \dot{C}_r$ is included using the algorithm given in Section 8.4.1 in order to let the system evolve towards the ground state.

Check A reasonable value for the time step is 0.1 (in atomic units) and for the frictional constant γ the value 1 (in atomic units) is chosen. In Figure 9.2, the energy as a function of the ‘time’ is shown. It is seen that for a nuclear separation

of $X = 1$ the energy tends to $-2.078\,547\,6$ a.u., the same value as was found in Problem 4.9.

9.3.2 The nuclear motion

In this section we describe the inclusion of the nuclear forces in the equations of motion and apply this to the vibration of the hydrogen molecule. Essentially, what we have to do is to calculate the derivative of the total energy with respect to the nuclear separation X . The results obtained using the Car–Parrinello HF method are exactly equivalent to those obtained by the force field method as we have a pair potential only; we describe it here only to illustrate the method. There are two contributions to this derivative. First of all, the energy contains a Coulomb interaction $1/X$ between the two nuclei and the electron Hamiltonian contains Coulomb attractions between the electrons and the nuclei, which depend on X . There is, however, yet another contribution from the dependence of the basis functions χ_r on X : remember the basis functions are centred on the nuclei, so varying the positions of the latter changes the matrix elements of the Fock matrix and the overlap matrix. In the following we shall not distinguish explicitly between all these contributions, but it is useful to know that contributions to the forces due to the variation of the basis functions with the nuclear positions are called *Pulay forces* [7]. If the basis functions do not depend on the nuclear coordinates, as is the case with plane wave basis sets, which are often used in conjunction with pseudopotentials, Pulay forces are absent. We shall now calculate the derivatives of the matrix elements of the Fock matrix and the overlap matrix with respect to the nuclear separation in the hydrogen molecule.

Expressions for the various matrix elements were given in Section 4.8. We use notations similar to those used in that section. The overlap matrix was given as

$$S_{\alpha,A;\beta,B} = \langle 1s, \alpha, A | 1s, \beta, B \rangle = \left(\frac{\pi}{\alpha + \beta} \right)^{3/2} \exp \left[-\frac{\alpha\beta}{\alpha + \beta} |\mathbf{R}_A - \mathbf{R}_B|^2 \right], \quad (9.33)$$

and we see that if both basis functions are centred on the same nucleus ($A = B$), this matrix element does not depend on X . For two basis functions centred on different nuclei, $|\mathbf{R}_A - \mathbf{R}_B| = X$, and we find

$$\frac{d}{dX} \langle 1s, \alpha, A | 1s, \beta, B \rangle = -2 \frac{\alpha\beta}{\alpha + \beta} X S_{\alpha,A;\beta,B}. \quad (9.34)$$

The matrix elements of the kinetic energy operator for two orbitals centred on the same atom are again independent of X , and for the elements between basis functions on different nuclei we have, using $\sigma = \alpha\beta/(\alpha + \beta)$ (see Section 4.8):

$$\langle 1s, \alpha, A | -\frac{1}{2} \nabla^2 | 1s, \beta, B \rangle = [3\sigma - 2\sigma^2 X^2] S_{\alpha,A;\beta,B}. \quad (9.35)$$

Taking the derivative with respect to X we find

$$\begin{aligned} \frac{d}{dX} \langle 1s, \alpha, A \left| -\frac{1}{2} \nabla^2 \right| 1s, \beta, B \rangle \\ = -4\sigma^2 X S_{\alpha,A;\beta,B} + [3\sigma - 2\sigma^2 X^2] \frac{d}{dX} S_{\alpha,A;\beta,B}. \end{aligned} \quad (9.36)$$

The Coulomb matrix element is given by

$$\langle 1s, \alpha, A \left| \sum_c \frac{1}{r_c} \right| 1s, \beta, B \rangle = \theta \sum_c S_{\alpha,A;\beta,B} F_0(t_c) \quad (9.37)$$

with $\theta = 2\sqrt{(\alpha + \beta)/\pi}$, $t_c = (\alpha + \beta)(PC)^2$ where P is the point

$$\mathbf{R}_P = \frac{\alpha \mathbf{R}_A + \beta \mathbf{R}_B}{\alpha + \beta}, \quad (9.38)$$

$PQ = \mathbf{R}_P - \mathbf{R}_Q$, and C is the position of the nucleus. The sum $\sum_c$ is over the two nuclei. F_0 was given in [Section 4.8](#) – its derivative is given by

$$F'_0(t) = \frac{e^{-t} - F_0(t)}{2t} \quad (9.39)$$

for $t \neq 0$, and $F'_0(0) = -1/3$. Taking the derivative, we obtain for two basis functions centred on the same nucleus:

$$\frac{d}{dX} \langle 1s, \alpha, A \left| \sum_c \frac{1}{r_c} \right| 1s, \beta, B \rangle = 2\theta S_{\alpha,A;\beta,B} F'_0(t) X(\alpha + \beta) \quad (9.40)$$

with $t = (\alpha + \beta)X$.

For basis functions centred on different nuclei, we have

$$\begin{aligned} \frac{d}{dX} \langle 1s, \alpha, A \left| \sum_c \frac{1}{r_c} \right| 1s, \beta, B \rangle &= \theta \frac{d}{dX} (S_{\alpha,A;\beta,B}) \sum_c [F_0(t_1) + F_0(t_2)] \\ &+ 2 \frac{\theta}{\alpha + \beta} S_{\alpha,A;\beta,B} [F'_0(t_1) \alpha^2 + F'_0(t_2) \beta^2] X. \end{aligned} \quad (9.41)$$

where

$$t_1 = \frac{\alpha^2 X^2}{\alpha + \beta}; \quad (9.42a)$$

$$t_2 = \frac{\beta^2 X^2}{\alpha + \beta}. \quad (9.42b)$$

Finally the four-electron matrix element is given by

$$\langle \alpha, A; \gamma, C | g | \beta, B; \delta, D \rangle = \rho S_{\alpha,A;\beta,B} S_{\gamma,C;\delta,D} F_0(t) \quad (9.43)$$

with

$$t = \frac{(\alpha + \beta)(\gamma + \delta)}{\alpha + \beta + \gamma + \delta} (PQ)^2, \quad (9.44)$$

with $\mathbf{R}_P$ as given above and

$$\mathbf{R}_Q = \frac{\gamma \mathbf{R}_C + \delta \mathbf{R}_D}{\gamma + \delta}, \quad (9.45)$$

and

$$\rho = 2 \sqrt{\frac{(\alpha + \beta)(\gamma + \delta)}{\pi(\alpha + \beta + \gamma + \delta)}}. \quad (9.46)$$

From this form it follows directly that

$$\begin{aligned} \frac{d}{dX} \langle \alpha, A; \gamma, C | g | \beta, B; \delta, D \rangle &= \rho \left(\frac{d}{dX} S_{\alpha, A; \beta, B} \right) S_{\gamma, C; \delta, D} F_0(t) \\ &+ \rho S_{\alpha, A; \beta, B} \left(\frac{d}{dX} S_{\gamma, C; \delta, D} \right) F_0(t) \\ &+ \rho S_{\alpha, A; \beta, B} S_{\gamma, C; \delta, D} F'_0(t) \frac{(\alpha + \beta)(\gamma + \delta)}{\alpha + \beta + \gamma + \delta} \frac{2(PQ)^2}{X} \end{aligned} \quad (9.47)$$

where we have used the fact that (PQ) is proportional to X in order to obtain the last term on the right hand side.

Using these matrix elements, it is possible to construct the derivatives of the Fock matrix and of the overlap matrix with respect to X , and this gives the force on X which is needed in the Verlet algorithm. Note that the nuclear kinetic energy is given by

$$E_{\text{kin,nucl}} = \frac{M_n}{2} \left[\left(\frac{\dot{X}}{2} \right)^2 + \left(\frac{\dot{X}}{2} \right)^2 \right] = \frac{M_n}{4} \dot{X}^2. \quad (9.48)$$

Therefore, in the equation of motion for X , half the proton mass (that is, the reduced mass of the two nuclei) has to be used.

Only the ratio of the masses occurring in the electronic and nuclear kinetic energy is relevant – changing the time step h corresponds to an overall rescaling of the masses. In fact, because the mass occurs in the equation of motion in combination with an acceleration (or, in the kinetic energy, with a velocity squared), rescaling the mass by a factor b and time with a factor $\sqrt{b}$ does not change the calculated motion.

PROGRAMMING EXERCISE

Extend the program of the previous subsection to include the nuclear motion.

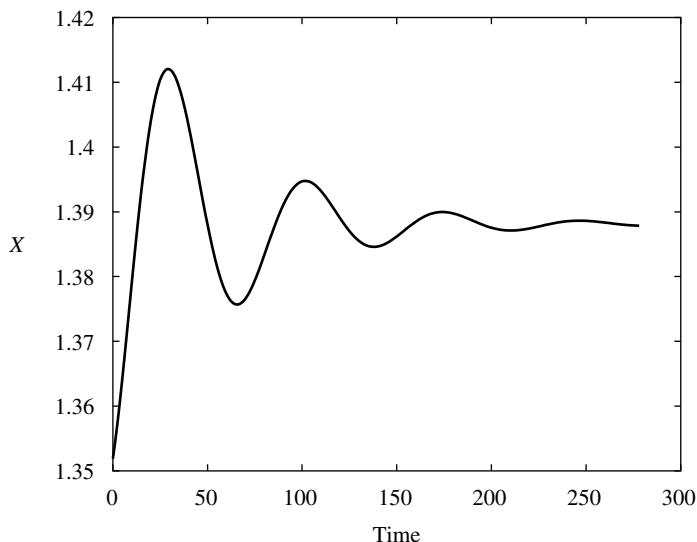


Figure 9.3. The change of the separation X between the nuclei of a hydrogen molecule as a function of time. The number of nuclear integration steps is shown along the X -axis. The nuclear integration step size is 4.3 (in atomic units). The integration step for the electrons was 0.1. Twelve thousand electron integration steps were carried out. The electrons experience a friction with damping constant $\gamma = 1$, and the nuclei are damped with a friction constant of 5.

Check 1 Take the nuclear mass e.g. 1000 times larger than the electron mass. The nuclei will move very slowly in comparison with the electrons because they are so much heavier. If friction is included, the nuclei should end up with zero velocity at their equilibrium spacing, which is at $X = 1.3881a_0$ (within the HF approximation and using exclusively s-basis functions). This is to be compared with the experimental value of $1.401a_0$. The behaviour of X as a function of time is shown in Figure 9.3.

Check 2 If friction is not included, the nuclei will oscillate around their equilibrium separation. Use 1836.15 for the proton mass. The frequency for an initial separation of 1.35 Bohr radii is found to be 13.5×10^{13} Hz, to be compared with the value 13.64×10^{13} Hz obtained above from fitting a parabola to the bottom of the effective potential well in Figure 9.1, and with the experimental value, which is 12.48×10^{13} Hz. The parabola was characterised by a ‘spring constant’ $\kappa = 0.385$. The behaviour of X as a function of time is shown in Figure 9.4. Check that the results in Figure 9.4 comply with this value (note that the time step in this figure is 4.3 in reduced units).

It is possible and advisable to use fewer integration steps for the nuclear equation of motion than for the electronic one: the nuclei move much more slowly than the

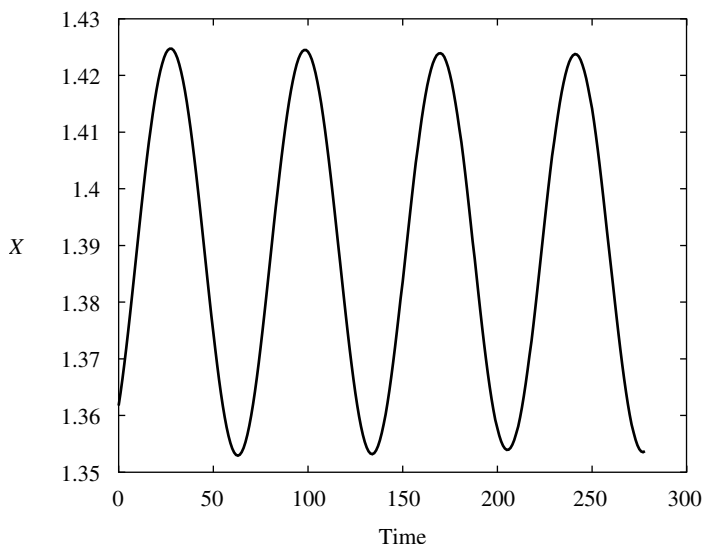


Figure 9.4. The change of the separation X between the nuclei of a hydrogen molecule as a function of time. The number of nuclear integration steps is shown along the X -axis. The nuclear integration step size is 4.3 (in atomic units). The integration step for the electrons was 0.1. Twelve thousand electron integration steps were carried out. The electrons experience a friction with damping constant $\gamma = 1$ during the first 4000 steps; the nuclei experience no friction.

electrons, and a nuclear displacement is computationally expensive because the overlap, Hamilton and Fock matrices have to be calculated again. As the nuclei are moving much more slowly than the electrons this does not affect the overall accuracy significantly, provided the number of electronic integration steps carried out between two nuclear displacements is smaller than $\mathcal{O}(\sqrt{M_n/\mu})$ (see also above).

9.4 Orthonormalisation; conjugate gradient and RM-DIIS techniques

In the previous sections, we have discussed the ‘bare-bones’ Car–Parrinello method and applied it to a simple system. There is much more to it – quantum molecular dynamics is still a very active field within computational condensed matter research – and the interested reader is referred to the review papers by Payne *et al.* [8], and Marx and Hutter [9] for details. In this section we describe some elements of the Car–Parrinello method in more detail, and briefly describe a variant of it, using conjugate gradients (see [Appendix A4](#)) for minimising the electronic energy.

9.4.1 Orthogonalisation of the electronic orbitals

The orthogonalisation of the electronic orbitals is maintained through the Lagrange multipliers Λ_{kl} , whose values therefore vary with time. The procedure to calculate

these values depends on the particular integration algorithm used, which is usually the Verlet algorithm. In the previous section you have seen how this is done in the case of one orbital, where only the normalisation matters – for more orbitals, understanding the different procedures is quite subtle.

In the following we shall use the notation

$$\varepsilon = \sum_k \langle \psi_k | H | \psi_k \rangle \quad (9.49)$$

for the total energy for a set of orthonormal orbitals ψ_k . H stands for the Fock matrix in HF, and in DFT it is the Kohn–Sham Hamiltonian. Let us write down the Verlet equations of motion for the electronic orbitals:

$$\psi_k(t+h) = 2\psi_k(t) - \psi_k(t-h) - \frac{2h^2}{\mu} (H\psi_k - \sum_l \Lambda_{kl}\psi_l). \quad (9.50)$$

The – as yet unknown – multipliers Λ_{kl} are symmetric, $\Lambda_{kl} = \Lambda_{lk}$, and therefore represent $N(N+1)/2$ independent values, which are determined by the $N(N+1)/2$ orthonormality conditions. Hence the Lagrange multipliers are uniquely defined. It might therefore be surprising that several different orthogonalisation algorithms exist [8, 10]. The reason is that a unitary transformation of the set of orbitals leaves the set orthonormal: the set $\{\psi'_k\}$ defined by

$$\psi'_k = \sum_l U_{kl}\psi_l \quad (9.51)$$

is orthonormal. Moreover, a unitary transformation leaves the charge density unchanged – remember the DFT energy depends on the density and not on the individual orbitals. Also, the Slater determinants forming the basis functions in the Hartree–Fock theory are invariant under unitary transformations (see Problem 4.7). It should be noted that such a transformation of the set ψ_k is accompanied by a similarity transform of the Lagrange parameters:

$$\Lambda'_{kl} = \sum_{mn} U_{km}^\dagger \Lambda_{mn} U_{nl} \quad (9.52)$$

as can be verified directly from the equation of motion (9.21). Different orthonormalisation algorithms result in sets of orbitals which span the same space of functions but which are slightly rotated with respect to each other.

Such a rotation may have a tremendous effect on the performance of the Verlet algorithm. To see this, consider a permutation of the orbitals (which is a special case of a unitary transformation), carried out between two time steps. This permutation does not affect the density but it may have a disastrous effect on the integration of the equations of motion: the (fictitious) velocities of the permuted orbitals increase suddenly to values of $\mathcal{O}(h^{-1})$, because the permutation disrupts

the smooth evolution of the orbitals! However, if the rotation is always close to the unit transformation,

$$U = 1 + h^2 A \quad (9.53)$$

where A is a Hermitian transformation of order one, varying smoothly with time, the Verlet algorithm will still work: apart from the motion governed by the equation of motion, the algorithm might induce some extra forces which cause the orbitals to rotate smoothly in Hilbert space, and this latter motion can be dealt with perfectly by the Verlet algorithm. It is difficult to see whether orthogonalisation algorithms satisfy these requirements and it is therefore easiest to construct the algorithm such that it is equivalent to the unambiguous time evolution resulting from the Verlet algorithm (without extra rotation) to a precision of at least order h^4 , which is the overall precision of the Verlet algorithm.

A method which is based on the Verlet algorithm and which solves the Λ_{kl} in (9.50) by the orthogonality requirements is the iterative algorithm called SHAKE by Ryckaert *et al.* [5], which was mentioned in Section 8.6.3. This algorithm was used in the original work of Car and Parrinello [11]. It is straightforward and does not introduce rotations of the set of orbitals. Moreover, it orthogonalises the states to arbitrary precision (depending on the number of iterations performed). For details we refer to the cited literature.

Most other methods first predict the form of the (orthonormal) ψ_k at the next time step with some precision and then perform an additional orthonormalisation of these predicted orbitals by constructing orthonormal linear combinations of them. The idea behind this is that if the prediction is accurate, only a few orthonormalisation iterations are needed. As the Verlet algorithm prescribes an orthonormalisation by mixing in the $\psi_k(t)$ through the Lagrange multipliers (see Eq. (9.50)), and not the $\psi_k(t + h)$, such a final re-orthonormalisation can only be justified if the changes involved are of order h^4 which is the overall accuracy of the Verlet algorithm for a single step. Therefore these algorithms must first predict the new values to order $\mathcal{O}(h^4)$ and the re-orthonormalisation should yield the new states lying close to the old states. Note that after each step orthonormality is of then satisfied to machine precision whereas the error in the integration algorithm is of order h^4 .

Let us now consider one such algorithm in detail. Over a time step h , the orbitals ψ_k shift by an amount of order h owing to the fact that they have a velocity. The acceleration, which is caused by the force due to the derivative of the total energy and by the force due to the Lagrange multiplier terms, then gives an additional shift of order h^2 . The Hamiltonian force term occurring in the Car–Parrinello equations of motion is given by $-H\psi_k$. First we neglect this force term for simplicity and find Lagrange multipliers which guarantee orthonormality in the absence of forces.

The equation of motion then reads

$$\mu \ddot{\psi}_k = \sum_l \Lambda_{kl} \psi_l \quad (9.54)$$

and the second derivative of the constraint equations gives

$$\langle \ddot{\psi}_k | \psi_l \rangle + \langle \psi_k | \ddot{\psi}_l \rangle + 2 \langle \dot{\psi}_k | \dot{\psi}_l \rangle = 0, \quad (9.55)$$

so that, using the orthonormality of the orbitals, we find

$$\Lambda_{kl} = -\mu \langle \dot{\psi}_k | \dot{\psi}_l \rangle. \quad (9.56)$$

Of course we do not know the exact time derivative of the wave function, but we can use the Taylor expansion

$$\dot{\psi}_k(t) = \frac{\psi_k(t) - \psi_k(t-h)}{h} + \frac{h}{2} \ddot{\psi}_k(t) + \mathcal{O}(h^2), \quad (9.57)$$

and take for the approximate time derivative of the wave function at time t :

$$\dot{\tilde{\psi}}_k(t) = \frac{\psi_k(t) - \psi_k(t-h)}{h}. \quad (9.58)$$

From the time derivative of the constraint equation it follows that

$$\left\langle \frac{d\psi_k}{dt} \middle| \psi_l \right\rangle = 0, \quad \text{all } k \text{ and } l \quad (9.59)$$

and using this result, it is easy to show by a Taylor expansion of the $\psi_k(t-h)$ that

$$\langle \dot{\tilde{\psi}}_k(t) | \dot{\tilde{\psi}}_l(t) \rangle = \langle \dot{\psi}_k(t) | \dot{\psi}_l(t) \rangle + \mathcal{O}(h^2). \quad (9.60)$$

As the Lagrange multipliers occur in the Verlet algorithm with a factor h^2 in front of it, the accuracy of order h^2 with which we obtain their values by using this procedure is sufficient (that is, $\mathcal{O}(h^4)$).

Next we include the Hamiltonian force $-H\psi_k$ into the problem. A similar analysis as above (Eqs. (9.54–9.56)) leads to the following equation for the Lagrange multipliers:

$$\Lambda_{kl} = 2 \langle \psi_k | H | \psi_l \rangle - \mu \langle \dot{\psi}_k | \dot{\psi}_l \rangle. \quad (9.61)$$

The term $\langle \psi_k | H | \psi_l \rangle$ can easily be calculated in the program. If we use these Lagrange parameters in the Verlet algorithm we obtain a set of orbitals which is accurate to order h^4 and is therefore in particular orthonormal to order h^4 . Next we must apply an orthonormalisation algorithm which leaves the orthonormal set close to the near-orthonormal one. One possibility is an iterative algorithm used by Car and Parrinello [12]:

$$\psi'_k = \psi_k - \frac{1}{2} \sum_{l \neq k} \langle \psi_l | \psi_k \rangle \psi_l \quad (9.62)$$

which is repeated until the orbitals do not change any more. By inspection of this algorithm it is seen that if the orbitals are orthonormal up to order h^4 , they will change to an extent within that order over a time step. The same holds for the Gram–Schmidt orthonormalisation, which is given by the algorithm

$$\psi'_k = \psi_k - \sum_{l < k} \langle \psi_l | \psi_k \rangle \psi_l. \quad (9.63)$$

In contrast to the previous algorithm, the Gram–Schmidt algorithm depends on the way in which the orbitals are ordered. In particular, the orbital which we take as the first in the Gram–Schmidt process remains unchanged. It is clear from the foregoing analysis that this does not really matter if the states are already orthogonal to order h^4 .

Another possible way to orthogonalise the orbitals is to calculate the Lagrange parameters such that the final orbitals will be orthonormal. For example, we first calculate the new orbitals without taking the Lagrange multipliers into account:

$$|\tilde{\psi}_k(t+h)\rangle = 2|\psi_k(t)\rangle - |\psi_k(t-h)\rangle - \frac{2h^2}{\mu} H |\psi_k(t)\rangle, \quad (9.64)$$

and then we calculate the Lagrange parameters X_{ij} such that the orbitals

$$|\psi_k(t+h)\rangle = |\tilde{\psi}_k(t+h)\rangle + \sum_l X_{kl} |\psi_l(t)\rangle \quad (9.65)$$

form an orthonormal set. Obviously, the X_{kl} are related to the Λ_{kl} by

$$X_{kl} = \frac{2h^2}{\mu} \Lambda_{kl}. \quad (9.66)$$

The parameters X_{ij} should satisfy a matrix equation which can conveniently be formulated after introducing the matrices $A_{kl} = \langle \tilde{\psi}_k(t+h) | \tilde{\psi}_l(t+h) \rangle$ and $B_{kl} = \langle \psi_k(t) | \tilde{\psi}_l(t+h) \rangle$ as:

$$\mathbf{X}\mathbf{X}^\dagger + \mathbf{X}\mathbf{B} + \mathbf{B}^\dagger\mathbf{X}^\dagger = \mathbf{I} - \mathbf{A}. \quad (9.67)$$

This can be solved iteratively by the straightforward reformulation

$$\mathbf{X}^{(n+1)} = \frac{1}{2}[\mathbf{I} - \mathbf{A} + \mathbf{X}^{(n)}(\mathbf{I} - \mathbf{B}) + (\mathbf{I} - \mathbf{B}^\dagger)\mathbf{X}^{(n)} - \mathbf{X}^{(n)}\mathbf{X}^{(n)\dagger}]. \quad (9.68)$$

As an initial guess, we take $\mathbf{X} = \frac{1}{2}(\mathbf{I} - \mathbf{A})$, which is close to the first guess for Λ_{kl} found above.

Summarising the integration algorithm so far:

We have $|\psi_k(t)\rangle$ and $|\psi_k(t-h)\rangle$.

Find $|\tilde{\psi}_k(t)\rangle = (|\psi_k(t)\rangle - |\psi_k(t-h)\rangle)/h$.

Now we may proceed in two ways:

- (i) Calculate as a first guess $\Lambda_{kl} = 2\langle\psi_k(t)|H|\psi_l(t)\rangle - \mu\langle\dot{\psi}_k(t)|\dot{\psi}_l(t)\rangle$ and calculate $|\psi(t+h)\rangle$ by the algorithm (9.50). Then refine the solution thus obtained by orthonormalisation.
- (ii) Calculate the new solution by the algorithm (9.50) *without* the constraint term. Then calculate the Lagrange parameters X_{ij} by the above algorithm. The constraint term is then added to the solution.

It is possible to turn the Verlet algorithm into a velocity Verlet form. In that case, not only the new orbitals are orthonormalised, but also the constraint $\langle\dot{\psi}_k(t)|\psi_l(t)\rangle$ is satisfied rigorously at all times. The algorithm requires a bit more work and storage. Its advantage is that it can be extended more easily within a Nosé–Hoover thermostat scheme – for details see Refs. [9, 13]. We give the algorithm here for completeness:

$$|\dot{\tilde{\psi}}_k(t+h)\rangle = |\dot{\psi}_k(t)\rangle + \frac{h}{\mu}H|\psi_k(t)\rangle; \quad (9.69a)$$

$$|\tilde{\psi}_k(t+h)\rangle = |\psi_k(t)\rangle + h|\dot{\tilde{\psi}}_k(t+h)\rangle; \quad (9.69b)$$

$$|\psi_k(t+h)\rangle = |\tilde{\psi}_k(t+h)\rangle + \sum_{ij} X_{ij}|\psi_k(t)\rangle; \quad (9.69c)$$

$$|\dot{\psi}'_k(t+h)\rangle = |\dot{\tilde{\psi}}_k(t+h)\rangle + \frac{h}{\mu}H|\psi_k(t+h)\rangle; \quad (9.69d)$$

$$|\psi_k(t+h)\rangle = |\dot{\psi}'_k(t+h)\rangle + \sum_{ij} Y_{ij}|\psi_k(t+h)\rangle. \quad (9.69e)$$

The matrix X_{ij} is the same as used in the standard Verlet procedure above; the matrix Y_{ij} is simply calculated in terms of the matrix $C_{kl} = \langle\psi_k(t+h)|\dot{\psi}'_l(t+h)\rangle$ as

$$Y_{kl} = -\frac{C_{kl} + C_{kl}^\dagger}{2}. \quad (9.70)$$

If we include friction in the equations of motion, we are allowed more freedom, as the only requirement is that the orbits will become stationary by some damping mechanism. In this case, one can take the Λ_{kl} to be diagonal:

$$\Lambda_{kl} = \varepsilon_k \delta_{kl} \quad (9.71)$$

and the equation of motion for the stationary state leads directly to

$$\varepsilon_k = \langle\psi_k|H|\psi_k\rangle, \quad (9.72)$$

and this form of the Lagrange parameters is then used throughout the simulation, that is, even when if the orbitals move. However, this form of the Lagrange multipliers

does not preserve orthonormality if we are not at the energy minimum; therefore, in the simulation, we use these values for the ε_k and perform a Gram–Schmidt orthonormalisation afterwards in order to prevent the orbitals all evolving to the ground state of the Kohn–Sham or Fock equation. The orbitals will then tend to the eigenvalues of the Hamiltonian, as these are the stationary solutions of the equations of motion.

9.4.2 The conjugate gradient method

The Car–Parrinello technique is based upon three ideas. First, the forces are calculated during the simulation, and only for those nuclear configurations which are actually visited. Second, the electronic structure can be determined by minimising the energy using an arbitrary minimisation method, that is, not necessarily by a self-consistency iteration as in [Chapters 4 and 5](#). Car and Parrinello choose the molecular dynamics method for this purpose. Finally, we might pay a price in accuracy by not requiring the electrons to relax to the minimum energy state before each nuclear displacement.

In the conjugate gradients approach, the treatment of the electronic degrees of freedom differs from that in MD simulation methods. The idea is that if we abandon the usual self-consistency iterations which in the conventional HF and DFT approaches lead to the minimum of the electronic energy, we might as well apply any efficient minimisation method to the electronic energy – for example the conjugate gradients method – see [Section A4](#). Using the conjugate gradients method enables us to keep the electronic degrees of freedom much closer to the ground state than in the Car–Parrinello method.

The conjugate gradients technique enables us to calculate a local minimum of an arbitrary smooth function depending on a number of variables. In fact, in our case we must perform the minimisation with a constraint. That is, we must minimise the energy, $E[n]$, as a function of the orbitals, ψ_k , using the gradient of the function

$$E[n] - \sum_{kl} \Lambda_{kl} \langle \psi_k | \psi_l \rangle, \quad (9.73)$$

where Λ_{kl} must be such that orthonormalisation is always ensured. Using the notation of the previous subsection, $\partial E / \partial \psi_k = H | \psi_k \rangle$, we find that the orthonormality condition leads to

$$\Lambda_{kl} = \langle \psi_k | H | \psi_l \rangle; \quad (9.74)$$

see Exercise 9.3. In the conjugate gradients method we need the steepest descent direction, which is the opposite of the gradient of the function (9.73). Note that H depends on the orbitals ψ_k . Neglecting this dependence, we obtain for the steepest

descent direction:

$$\zeta_k = -2 \left[H\psi_k - \sum_l \langle \psi_k | H | \psi_l \rangle \psi_l \right]. \quad (9.75)$$

In a method proposed by Stich *et al.* [14], this approximation is used in the line minimisations. Gillan [15] has devised a conjugate gradients method in which this approximation is not made. Finally, Teter *et al.* [16] have proposed a particularly efficient method in which only one state ψ_k is updated at a time. See Problems 9.3 and 9.5 for examples.

The conjugate gradients method is known to converge slowly for this problem. The reason is that the steepest descent direction, which should be close to the difference between the current orbitals $\{\psi_k\}$ and those that minimise the energy, $\{\tilde{\psi}_k\}$, may not have this property. To see what goes wrong, let us expand $\delta\psi_k = \psi_k - \tilde{\psi}_k$ in the eigenstates ξ_p of the Hamiltonian:

$$\delta\psi_k = \sum_p \alpha_{pk} \xi_p. \quad (9.76)$$

Note that whereas the indices k and l in the previous discussion run only over the occupied electronic states, the index p runs over *all* the eigenstates of the variational Hamiltonian matrix, and their number is equal to the number of states in the variational basis set used. Working out the steepest descent direction for this state, we have

$$\zeta_k = -2 \sum_p (\alpha_{pk} \varepsilon_p - \sum_l \Lambda_{kl} \alpha_{lk}) \xi_p \quad (9.77)$$

(note that the part involving $\tilde{\psi}_k$ on the right hand side vanishes). The right hand side will contain important contributions from the high energy levels, and they will spoil the desired proportionality between steepest descent direction and $\delta\psi_k$. This can be remedied by an extension to the method, called *pre-conditioning* [17]. We shall not treat this in detail here – for details see Refs. [8, 16, 17].

The conjugate gradients method can be readily applied to energy minimisation problems. However, if we want to perform a dynamic simulation for the atoms, that is, without friction acting on the nuclei, a rather subtle problem arises. To see this, consider a nucleus which is moving in the positive x -direction with an electronic charge distribution around it, which has converged to the ground state. As the nucleus moves, the charge cloud starts lagging behind and this discrepancy might grow larger and larger with time. But this is not possible in the Verlet algorithm as this symplectic algorithm does not allow for energy drift (see Section 8.4). In the Verlet simulation, the electron charge cloud keeps oscillating around the nucleus so that errors remain bounded, and even cancel out on average. This is an important advantage of the molecular dynamics approach of Car and Parrinello [18–20]. There is, however, the possibility that the electron system absorbs energy from the nuclear

system and that the negative charge cloud would therefore oscillate more and more violently around the nuclei. To avoid this, the total simulation time should be kept short, or the electrons and the nuclei should both be coupled to a Nosé–Hoover thermostat (see [Section 8.5.1](#)), perhaps at different temperatures [\[21\]](#).

In the conjugate gradients method, the errors are more erratic than oscillatory in nature. Therefore this cancellation effect will not occur in that case, and it is necessary to keep the electronic charge distribution very close to the energy minimum at any time in order to avoid an unstable propagation of the errors spoiling the results. This means that we must perform many conjugate gradient steps, which slows the calculation down considerably. The only way to achieve good performance using the conjugate gradient method is by extrapolating the orbitals at the next time step from the previous ones, so that the conjugate gradients iterations start off from a configuration which is close to the exact energy minimum [\[22\]](#).

Details concerning the conjugate gradients method can be found in [Refs. \[8, 14, 16, 17\]](#).

9.4.3 The RM-DIIS technique

Another widely used technique for finding the optimal orbitals is the RM-DIIS technique. The abbreviation stands for residual minimisation by direct inversion of the iterative subspace. This can be applied to any problem in which a set of orthonormal eigenstates of a large Hamiltonian must be found. We describe the method briefly – for details see [Refs. \[9, 23, 24\]](#).

The eigenfunctions that we seek satisfy

$$(H - \varepsilon_n)|\psi_n\rangle = 0; \quad n = 1, \dots, N. \quad (9.78)$$

We now want to quantify the deviation of some approximate set of states from this. To this end we reformulate [\(9.78\)](#) as

$$|\psi_n\rangle = |\psi_n\rangle + \frac{1}{H_{nn}}(H - H_{nn})|\psi_n\rangle, \quad (9.79)$$

with

$$H_{nn} = \frac{\langle\psi_n|H|\psi_n\rangle}{\langle\psi_n|\psi_n\rangle}. \quad (9.80)$$

For the eigenstates we obviously have $\varepsilon_n = H_{nn}$. This form may seem a bit arbitrary but it naturally leads to an iterative scheme:

$$|\psi_n^{(j+1)}\rangle = |\psi_n^{(j)}\rangle + \frac{1}{H_{nn}}(H - H_{nn})|\psi_n^{(j)}\rangle. \quad (9.81)$$

The reason the correction term has a prefactor $1/H_{nn}$ is that in this way all eigenvectors have a similar correction, independent of the energy eigenvalue. The main

feature of Eq. (9.81) for us is that the correction factor can be used as an error estimate, which we call $|\Delta_j\rangle$:

$$|\Delta_j\rangle = \frac{1}{H_{nn}}(H - H_{nn})|\psi_n^{(j)}\rangle. \quad (9.82)$$

Suppose we have a sequence of estimates $|\psi_n^{(j)}\rangle, j = 1, \dots, J$ for the optimal states, which are not necessarily constructed according to the recipe of Eq. (9.81). From this sequence we construct a new state as a linear combination of the previous ones

$$|\psi_n^{J+1}\rangle = \sum_{j=1}^J d_j |\psi_n^{(j)}\rangle \quad (9.83)$$

such that it has a minimal error. We must, however, beware of the fact that the states can be scaled at will – upon rescaling they still satisfy the same linear equations. However, rescaling affects the norm of the error which we want to minimise. In order to have an unambiguous measure of the error we require that

$$\sum_{j=1}^J d_j = 1, \quad (9.84)$$

which turns out to be a convenient choice.

Substituting the linear expansion (9.83) in the expression for the error $|\Delta_{J+1}\rangle$ of the new state $|\psi_n^{J+1}\rangle$, we see that yields a linear combination of the individual errors $|\Delta_j\rangle$ of the $|\psi_n^{(j)}\rangle$:

$$|\Delta_{J+1}\rangle = \sum_{j=1}^J d_j |\Delta_j\rangle. \quad (9.85)$$

Now we shall be specific about the minimalisation of the norm of the error. This is given by

$$|\Delta_{J+1}|^2 = \sum_{j,k=1}^J d_j d_k \langle \Delta_j | \Delta_k \rangle. \quad (9.86)$$

We abbreviate the matrix elements $\langle \Delta_j | \Delta_k \rangle$ by a_{jk} . Minimising the error norm, respecting the appropriate constraint, which is included through a Lagrange parameter λ , leads to

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & 1 \\ a_{21} & a_{22} & \dots & a_{2n} & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} & 1 \\ 1 & 1 & \dots & 10 & \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \\ -\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}. \quad (9.87)$$

The resulting states $|\psi_n^J\rangle$ are not yet orthogonal; an orthogonalisation of the states must be performed after each of the steps described above.

9.4.4 Large systems

Periodic or other type of boundary conditions are not imposed a priori in QMD simulations. Boundary conditions may, however, be introduced by the basis set, which might consist of periodic functions, or functions vanishing at some boundary. There is an increasing interest in nonperiodic systems containing large numbers of atoms; these systems are said to be *mesoscopic*. Examples of mesoscopic systems and phenomena are the scanning tunnelling microscope tip and surface, grain boundaries, quantum dots and wires, biological macromolecules, etc. Often these systems are periodically continued; in the case of grain boundaries, for example, we might consider a system cell with a linear size corresponding to 10 atoms, and containing a grain boundary. Imposing periodic boundary conditions means that we are considering a system containing a periodic array of such grain boundaries.

If the system cell contains large numbers of atoms, it becomes very important to use a method for solving electronic structure and dynamics for which the computer time scales favourably with time. In this subsection we analyse the scaling behaviour for a few methods. Two parameters are important: first, the number N_{at} of atoms in the system, and second, the size N_{B} of the basis set. Of course, these numbers are not independent: they are usually proportional to each other. However, as the number of basis states exceeds the number of atoms sometimes by a factor of 100 or more, it is important to distinguish between the two in the time scaling.

It depends strongly on the type of basis functions used how the time required by a quantum molecular dynamics simulation scales with the number of atoms. In the case of plane wave basis sets, FFT techniques (see [Appendix A9](#)) can be used to increase the efficiency of the calculations. The kinetic energy is diagonal in a plane wave basis set; for evaluating the potential energy the FFT transforms the states into real space where the potential energy is diagonal. The plane wave basis leads to a time scaling (complexity) for one integration step in the quantum molecular dynamics method of $N_{\text{at}}N_{\text{B}} \ln N_{\text{B}}$.

Other methods use localised orbitals as basis functions so that the Hamiltonian couples only orbitals on neighbouring atoms – hence the Hamiltonian becomes sparse, and sparse matrix methods can be very efficient, giving essentially a scaling behaviour of $N_{\text{at}}N_{\text{B}}$. Obviously, this can be used in the calculation of the electronic structure using the conventional self-consistent approach where the Hamilton matrix has to be diagonalised several times. Using localised basis functions, the *recursion method* of Haydock [25–27] is particularly useful as it allows for calculating the density in a number of steps independent of N_{at} , and as long as the

nuclear positions remain fixed, the method scales as N_B , but with a large pre-factor. The method converges much faster for insulators and semiconductors than for metals [27]. Finally, discretising the degrees of freedom on a cubic lattice can lead to methods with a favourable time scaling [28, 29].

*9.5 Implementation of the Car–Parrinello technique for pseudopotential DFT

In this section we describe how the self-consistent pseudopotential code described in Section 6.7.3 can be extended to include Car–Parrinello dynamics. In the code of Section 6.7.3, we constructed the DFT Hamiltonian, diagonalised it and calculated the density which was then used to construct a new Hamiltonian, and so on until convergence was achieved. We then calculated the total energy from the orbitals, the density and the nuclear positions (which were taken to be fixed).

Now we want to use the derivative of the total energy with respect to the orbital degrees of freedom *and* with respect to the nuclear positions in order to formulate equations of motions which can then be solved using the (velocity) Verlet algorithm. The rather complicated expression for the energy leads to even more complicated expressions for these forces. We start with the orbital forces. They follow directly from the expression for the total energy given in Section 6.7.7. If we calculate the orbital forces from this energy, we obtain by direct differentiation: [9]

$$\begin{aligned} \frac{\partial E_{\text{total}}}{\partial c_j^*(\mathbf{K})} &= \frac{K^2}{2} c_j(\mathbf{K}) + \sum_{\mathbf{K}'} V_{\text{loc}}^*(\mathbf{K} - \mathbf{K}') c_j(\mathbf{K}') \\ &\quad + \sum_n \sum_{lm} F_{jlm}^n e^{-i\mathbf{K} \cdot \mathbf{R}_n} Y_{lm}(\hat{\mathbf{K}}) h_{lm}^n p_m^l(K), \end{aligned} \quad (9.88)$$

where $V_{\text{loc}}^{\text{all}}$ is the total local potential:

$$V_{\text{loc}}(\mathbf{K}) = \sum_n \Delta V_{\text{loc}}(\mathbf{K}) + V_{\text{xc}}(\mathbf{K}) + 4\pi \frac{n_{\text{tot}}(\mathbf{K})}{K^2}. \quad (9.89)$$

The potentials occurring in this expressions are given in Section 6.7.3.

If we implement this force in a Verlet algorithm in which we reduce the velocities at each time step such as to mimic frictional forces which bring us to the stationary energy minimum, the result should be equal to that obtained in the Kohn–Sham program. For two silicon atoms placed at a distance of 1.05 a.u. in a cell of size 8 a.u., the converged energy is $-21.235\,347\,29$ atomic units.

For the dynamics of the nuclei, we need the gradients of the energy with respect to the nuclear coordinates. These can be obtained directly from the expression for the total energy. The nuclear gradients are generated by the local and nonlocal

parts of the pseudopotential, and the electrostatic energy. The expressions are:

$$\nabla_{\mathbf{R}_n} E_{\text{local}} = -\Omega \sum_{\mathbf{K}} i\mathbf{K} V_{\text{local},n}(\mathbf{K}) e^{-i\mathbf{K} \cdot \mathbf{R}_n} n^*(\mathbf{K}) \quad (9.90)$$

$$\nabla_{\mathbf{R}_n} E_{\text{nonlocal}} = \sum_j f_j \sum_{l,m \in n} [(F_{jlm}^n)^* h_{lm}^n \nabla_{\mathbf{R}_n} F_{jlm}^n + \nabla_{\mathbf{R}_n} (F_{jlm}^n)^* h_{lm}^n F_{jlm}^n]; \quad (9.91)$$

$$\nabla_{\mathbf{R}_n} E_{\text{ES}} = -\Omega \sum_{\mathbf{K} \neq 0} i\mathbf{K} \frac{n_{\text{tot}}^*}{K^2} n_{\text{core}}^n(\mathbf{K}) e^{-i\mathbf{K} \cdot \mathbf{R}_n} + \nabla_{\mathbf{R}_n} E_{\text{ovrl}}, \quad (9.92)$$

where

$$\nabla_{\mathbf{R}_n} F_{lm}^n = -\frac{1}{\sqrt{\Omega}} \sum_{\mathbf{K}} i\mathbf{K} e^{-i\mathbf{K} \cdot \mathbf{R}_n} c_j^*(\mathbf{K}) Y_{lm}(\hat{\mathbf{K}}) p_m^l(K), \quad (9.93)$$

and

$$\begin{aligned} \nabla_{\mathbf{R}_n} E_{\text{ovrl}} = & \sum_{n'}' \sum_{\mathbf{L}} \left\{ \frac{Z_n Z_{n'}}{|\mathbf{R}_n - \mathbf{R}_{n'} - \mathbf{L}|^3} \operatorname{erfc} \left[\frac{|\mathbf{R}_n - \mathbf{R}_{n'} - \mathbf{L}|}{\sqrt{2(\xi_n^2 + \xi_{n'}^2)}} \right] \right. \\ & \left. + \frac{2}{\sqrt{\pi}} \frac{1}{\sqrt{\xi_n^2 + \xi_{n'}^2}} \frac{Z_n Z_{n'}}{|\mathbf{R}_n - \mathbf{R}_{n'} - \mathbf{L}|^2} \exp \left[\frac{|\mathbf{R}_n - \mathbf{R}_{n'} - \mathbf{L}|}{\sqrt{2(\xi_n^2 + \xi_{n'}^2)}} \right] \right\} \\ & \times (\mathbf{R}_n - \mathbf{R}_{n'} - \mathbf{L}). \end{aligned} \quad (9.94)$$

The full implementation of the Car–Parrinello is quite cumbersome. For a hydrogen dimer, you should find a result similar to that of [Figure 9.4](#). Note that this frequency depends on the nuclear mass. Similar results should be found for a silicon dimer.

Exercises

- 9.1 [C] The Car–Parrinello method can be used to find the minimum of any variational energy functional. We use it in this problem for finding the ground state of a particle in a one-dimensional, infinitely deep potential well. This problem was treated in [Chapter 3, Section 3.2.1](#). We use N variational basis functions of the form

$$\chi_r(x) = x^r (x - a)(x + a), \quad r = 0, 1, 2, \dots, N - 1$$

from which a variational state $\psi(x)$ is built as

$$\psi(x) = \sum_r C_r \chi_r(x).$$

The Euler–Lagrange equations for the Lagrangian with potential

$$E[C_r] = \sum_{rs} C_r C_s \langle \chi_r | H | \chi_s \rangle = \sum_{rs} C_r C_s H_{rs}$$

and the normalisation constraint

$$\langle \psi | \psi \rangle = \sum_{rs} C_r C_s \langle \chi_r | \chi_s \rangle = \sum_{rs} C_r C_s S_{rs}$$

are given by

$$\mu \ddot{C}_r = 2 \sum_s (H_{rs} C_s - \Lambda S_{rs} C_s).$$

We solve this equation of motion using the Verlet algorithm with friction. Note that Λ is determined by the normalisation condition. Therefore we first perform an integration step with $\Lambda = 0$ and then calculate Λ from (9.32). You are free to choose μ , the frictional constant and the time step h , although they are not independent.

The energy should converge to 2.467 40 as found in Section 3.2.1.

- 9.2 [C] Extend the program of the previous problem to include more than one state. Each state $\psi_k(x)$ has its own set of coefficients C_{rk} :

$$\psi_k(x) = \sum_r C_{rk} \chi_r(x).$$

We consider K states. There are now $K(K+1)/2$ constraint equations:

$$\langle \psi_k | \psi_l \rangle = \sum_{rs} C_{rk} S_{rs} C_{sl} = \delta_{kl}$$

(interchanging k and l gives the same equation). The Euler–Lagrange equations now become

$$\mu \ddot{C}_{kr} = \sum_l \sum_s (H_{rs} - \Lambda_{kl}) C_{sl}.$$

Because we include friction in the problem, we can take Λ_{kl} diagonal:

$$\Lambda_{kl} = \varepsilon_k \delta_{kl},$$

and at each step we estimate ε_k as

$$\varepsilon_k = \sum_{rs} C_{rk} C_{sk} H_{rs}$$

see Section 9.4.1.

Implement these equations in a computer program and compare your results with those presented in Section 3.2.1.

- 9.3 [C] Consider again the deep potential well of the previous two problems. We now use the conjugate gradients method for finding the eigenvalues, by using it to minimise the energy functional. It is assumed that you have a conjugate gradient routine available.

Let us first consider the ground state. This can be written as

$$\psi_G(x) = \sum_r C_r \chi_r(x),$$

where χ_r is the basis consisting of polynomials vanishing on the boundaries of the well. The energy functional

$$E[C_r] = \sum_{rs} C_r C_s H_{rs}$$

must be minimised subject to the normalisation constraint

$$\sum_{rs} C_r C_s S_{rs} = 1.$$

The Lagrangian function for this problem is

$$L[C_r] = \sum_{rs} C_r C_s H_{rs} - \lambda \sum_{rs} C_r C_s S_{rs}$$

where λ should be such that the normalisation remains guaranteed when moving in the steepest descent direction. The steepest descent direction ζ is given by

$$\zeta = \sum_r D_r \chi_r(x);$$

$$D_r = -2 \sum_s C_s H_{rs} + 2\lambda \sum_s C_s S_{rs}.$$

(a) Show that

$$\lambda = \langle \psi | H | \psi \rangle.$$

(b) Use this in applying the conjugate gradients method in order to find the ground state. Note that convergence is slow as no preconditioning is applied.

Now we consider the problem of finding more energy eigenstates ψ_k which are expanded in the basis set states as

$$\psi_k(x) = \sum_r C_{rk} \chi_r(x).$$

For N eigenstates we have $N(N+1)/2$ constraints.

The Lagrange function now reads

$$L[C_r] = \sum_k \sum_{rs} C_{rk} C_{sk} H_{rs} - \sum_{kl} \Lambda_{kl} \sum_{rs} C_{rk} C_{sl} S_{rs}.$$

(c) Show that

$$\Lambda_{kl} = \langle \psi_k | H | \psi_l \rangle.$$

(d) Use this form to find the four lowest eigenstates.

9.4 [C] Use the conjugate gradients technique for finding the minimum of the electronic energy of the hydrogen molecule with a fixed configuration of nuclei. This is a straightforward extension of the first program of the previous problem: there is only one normalisation constraint. Note however that the energy functional contains a term which is quartic in the C_r – see Eq. (9.26). Show that the steepest descent direction subject to the constraint is given by

$$D_r = -2 \sum_s F_{rs} C_s + 2\lambda \sum_s S_{rs} C_s$$

with

$$\lambda = \sum_{rs} C_r F_{rs} C_s.$$

Apply the conjugate gradients method to this problem and compare the results with the matrix diagonalisation method of [Chapter 4](#) and the molecular dynamics method of the present chapter.

References

- [1] G. M. Barrow, *Introduction to Molecular Spectroscopy*. New York, McGraw-Hill, 1962.
- [2] H. Hellmann, *Einführung in die Quantenchemie*. Leipzig, Deuticke, 1937.
- [3] R. P. Feynman, 'Forces in molecules,' *Phys. Rev.*, **56** (1939), 340–3.
- [4] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, 'Optimization by simulated annealing,' *Science*, **220** (1983), 671–80.
- [5] J. P. Ryckaert, G. Ciccotti, and H. J. C. Berendsen, 'Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes,' *J. Comput. Phys.*, **23** (1977), 327–41.
- [6] J. W. Negele, 'The mean-field theory of nuclear structure and dynamics,' *Rev. Mod. Phys.*, **54** (1982), 913–1015.
- [7] P. Pulay, 'Ab initio calculation of force constants and equilibrium geometries in polyatomic molecules. I. Theory,' *Mol. Phys.*, **17** (1969), 197–204.
- [8] M. C. Payne, M. P. Teter, D. C. Allen, T. A. Arias, and J. D. Joannopoulos, 'Iterative minimization techniques for ab initio total-energy calculations: molecular dynamics and conjugate gradients,' *Rev. Mod. Phys.*, **64** (1992), 1045–97.
- [9] D. Marx and J. Hutter, 'Ab initio molecular dynamics: theory and implementations,' in *Modern Methods and Algorithms of Quantum Chemistry, NIC series*, vol. 1. Jülich, John von Neumann Institute for Computing, 2000, pp. 301–449.
- [10] J. Q. Broughton and F. Khan, 'Accuracy of time-dependent properties in electronic structure calculations using a fictitious Lagrangian,' *Phys. Rev. B*, **40** (1989), 12098–104.
- [11] R. Car and M. Parrinello, 'Unified approach for molecular-dynamics and density-functional theory,' *Phys. Rev. Lett.*, **55** (1985), 2471–4.
- [12] R. Car and M. Parrinello, 'The unified approach for molecular dynamics and density functional theory,' in *Simple Molecular Systems at Very High Density* (A. Polian, P. Leboeury, and N. Boccard, eds.), *NATO Advanced Study Institute*, vol. 186, New York, Plenum, 1989, p. 455.
- [13] M. E. Tuckerman and M. Parrinello, 'Integrating the Car–Parrinello equations. I. Basic integration techniques,' *J. Chem. Phys.*, **101** (1994), 1302–15.
- [14] I. Stich, R. Car, M. Parrinello, and S. Baroni, 'Conjugate-gradient minimisation of the energy functional: a new method for electronic structure calculation,' *Phys. Rev. B*, **39** (1989), 4997–5004.
- [15] M. J. Gillan, 'Calculation of the vacancy formation energy in aluminium,' *J. Phys.–Cond Mat*, **1** (1989), 689–711.
- [16] M. P. Teter, M. C. Payne, and D. C. Allan, 'Solution of Schrödinger's equation for large systems,' *Phys. Rev. B*, **40** (1989), 12255–63.
- [17] P. E. Gill, W. Murray, and M. H. Wright, *Practical Optimization*. London, Academic Press, 1981.
- [18] M. C. Payne, 'Error cancellation in the molecular-dynamics method for total energy calculations,' *J. Phys.–Cond. Mat.*, **1** (1989), 2199–210.
- [19] R. Car, M. Parrinello, and M. C. Payne, 'Error cancellation in the molecular-dynamics method for total energy calculations: comment,' *J. Phys.–Cond. Mat.*, **3** (1991), 9539–43.
- [20] D. K. Remler and P. A. Madden, 'Molecular-dynamics without effective potentials via the Car–Parrinello approach,' *Mol. Phys.*, **70** (1990), 921–66.
- [21] P. Blochl and M. Parrinello, 'Adiabaticity in first-principles molecular dynamics,' *Phys. Rev. B*, **45** (1992), 9413–16.

- [22] T. A. Arias, J. D. Joannopoulos, and M. C. Payne, 'Ab initio molecular-dynamics techniques extended to large-length scale systems,' *Phys. Rev. B*, **45** (1992), 1538–49.
- [23] P. Pulay, 'Improved SCF convergence acceleration,' *J. Comp. Chem.*, **3** (1982), 556–60.
- [24] R. M. Martin, *Electronic Structure*. Cambridge, Cambridge University Press, 2004.
- [25] R. Haydock, V. Heine, and M. J. Kelly, 'Electronic structure based on the local atomic environment for tight-binding bands,' *J. Phys. C*, **5** (1972), 2845–58.
- [26] R. Haydock, 'The recursive solution of the Schrödinger equation,' in *Solid State Physics* (H. Ehrenreich, F. Seitz, and D. Turnbull, eds.), vol. 35, New York, Academic Press, 1980, pp. 216–94.
- [27] A. Gibson, R. Haydock, and J. P. LaFemina, 'Ab initio electronic structure computations with the recursion method,' *Phys. Rev. B*, **47** (1993), 6518–23.
- [28] S. Baroni and P. Giannozzi, 'Towards very large-scale electronic structure calculations,' *Europhys. Lett.*, **17** (1992), 547–52.
- [29] J. M. Thijssen and J. E. Inglesfield, 'Embedding muffin tins into a finite difference grid,' *Europhys. Lett.*, **27** (1994), 65–70.